Dysthe Evolution: Numerical Evidence and Existence

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Abstract

The recommended first project is a study of continued smooth evolution for the approved spatial free-surface water-wave Dysthe equation, using a qualified Fourier solver and a separately evaluated physics-informed neural network (PINN). The immediate output should be a reproducible map of resolved propagation intervals, derivative growth, and approximation error. An existence theorem requires additional control of stability and unresolved modes. We review Dysthe (1979) and Schober–Islas (2026, supplied version 1), identify two apparent linear-dispersion inconsistencies in the latter, and show why a real delayed mean flow is not automatically equivalent to its complex damping model. No new evolution campaign, trained PINN, or computer-assisted existence certificate is claimed here.

1 The research target

The question is whether prescribed smooth envelopes remain smooth as time or propagation distance increases. It is not primarily a search for solitary waves. Three tasks must be distinguished:

1. Construct a numerical trajectory and a PINN approximation for the same equation, initial data, domain, and interval.

2. Assess continued regularity by checking derivative norms, spectra, and the full equation residual under independent refinement.

3. Prove existence through a matching local theory and continuation estimate, or a rigorous a posteriori validation of a trajectory.

A successful first project can complete the first two tasks and prepare the third. A smooth plot, agreement between two methods, or a small training loss does not by itself complete the third.

Recommendation. Begin with `water-wave-dysthe-spatial-v1`, which already has a working reference solver. Use short propagation windows and a Fourier-in-profile PINN. Evaluate both methods using the full residual, including discarded frequencies. Target a rigorous certificate on one short periodic interval before attempting an all-distance claim. Keep the newer paper's temporal damping and relaxation constructions separately identified; do not insert them into the approved baseline by changing a coefficient.

The useful research result is: for a specified family of initial water-wave envelopes, how far are the approximations trustworthy, which derivatives grow first, what limits that interval, and can a rigorous bound certify at least one trajectory?

2 What the supplied papers establish

2.1 Dysthe (1979): a model and a stability calculation

Dysthe derives corrections beyond cubic nonlinear Schrödinger (NLS) theory from inviscid, irrotational, deep-water free-surface flow [1]. His A is a velocity-potential envelope. The full system, printed equations (2.19)–(2.21) with the mean-flow Laplace equation, evolves in time and includes two horizontal coordinates. It is not the project's spatial free-surface equation.

Slow modulation drives an induced mean potential. Its horizontal current changes the local carrier frequency and modifies sideband instability. Section 3 linearizes about a uniform wavetrain. For example, equation (3.10), printed p. 111, gives

$$\gamma_M = \tfrac{1}{2}(1 - 2A_0)A_0^2,$$

where A_0 is nondimensional carrier steepness. Figures 1–3 compare approximate instability predictions with previous calculations and measurements. These are not convergence tests of nonlinear trajectories.

The simplified equations (4.1)–(4.3), printed p. 113, retain the mechanism responsible for the stated leading stability correction. They should not replace the full evolution model in a regularity study without a separate justification. The paper supplies neither a well-posedness theorem nor a rigorous approximation-error theorem relative to the Euler equations.

Remark 2.1. Modulational instability means that certain small disturbances grow. It does not imply blow-up or nonexistence. Conversely, a smooth envelope solution need not stay within the weak-steepness, narrowband regime in which the water-wave approximation is physically credible.

2.2 Schober–Islas (2026): three distinct constructions

The supplied preprint [2] must be read as three constructions.

1. Its introductory equation (1.1) is a phenomenological temporal model with a complex coefficient in the nonlocal interaction:

$$iu_t + u_{xx} + 2|u|^2u + \epsilon\left[2u(1 + i\beta)\mathcal{H}\partial_x(|u|^2) - 8i|u|^2u_x + \tfrac{i}{2}u_{xxx}\right] = 0. \tag{2.1}$$

   The paper identifies $\beta = 0$ with a Hamiltonian Dysthe formulation. Its variable is not automatically the 1979 potential envelope or the approved spatial free-surface envelope.

2. Sections 2–6 use the Dias–Dyachenko–Zakharov (DDZ) potential-flow reduction with weak viscosity $\bar{\nu} = \epsilon^2\nu$. The reported fifth-order calculation produces derivative-dependent nonlocal terms but no explicit viscosity multiplier on the leading mean-flow operator. The sixth-order calculation isolates one carrier–mean contribution; it does not derive the complete sixth-order equation or the final net coefficient.

3. Section 7 introduces an independent relaxation closure for the mean flow. It is not derived from the DDZ hierarchy. Its adjustment time is prescribed, with physical origin and calibration unresolved.

The preprint does not derive a constant positive β from first principles and does not establish existence or continued smoothness. Its fifth-order cancellation is a conclusion of its formal expansion; we have not independently reproduced the full harmonic calculation. The paper itself distinguishes asymptotic order and operator structure from a term-by-term identification with its introductory equation.

2.3 Two apparent inconsistencies in the supplied version

The following findings were checked against the rendered PDF. They motivate a source/author audit, not an unannounced repair of the model.

Cubic dispersion. Equation (4.14), p. 9, prints

$$-\epsilon^2 \frac{8ik^3}{\omega^2} A_{XXX}.$$

For the stated dimensional $X = \epsilon x$, $T = \epsilon t$, $\omega^2 = gk$, and carrier $e^{i(\omega t - kx)}$, the coefficient is dimensionally inconsistent. Independently, put $A = e^{i(KX - \Omega T)}$ and remove viscosity and nonlinearity. Exact deep-water dispersion requires

$$\begin{aligned} \Omega &= \frac{\omega - \sqrt{g(k - \epsilon K)}}{\epsilon} \\ &= \frac{\omega K}{2k} + \epsilon \frac{\omega K^2}{8k^2} + \epsilon^2 \frac{\omega K^3}{16k^3} + \epsilon^3 \frac{5\omega K^4}{128k^4} + O(\epsilon^4). \end{aligned} \tag{2.2}$$

In the paper's normalization $2i\omega(A_T + \omega A_X/(2k)) + \cdots = 0$, the cubic coefficient consistent with this expansion is $-i\omega^2/(8k^3)$, not $-8ik^3/\omega^2$. Even at $k = \omega = 1$, their magnitudes differ by a factor of 64.

Missing quartic dispersion. Equations (5.6), (5.7), and (5.11), p. 12, claim a complete fifth-order condition, but their inviscid linearization contains no A_{XXXX}. In the same variables, (2.2) requires $-5\epsilon^3\omega^2 A_{XXXX}/(64k^4)$. No exact-dispersion operator or documented change of envelope variable explaining this absence was identified.

These are apparent inconsistencies in the supplied version, not a rederivation of its nonlinear hierarchy. Check possible corrections and the complete linear DDZ dispersion before implementing the higher-order equation. Correcting two terms would not validate all remaining coefficients.

3 Real mean-flow delay is not automatically nonlinear damping

Adopt $\widehat{\mathcal{H}f}(k) = -i\,\mathrm{sgn}(k)\widehat{f}(k)$, so $\mathcal{H}\partial_x = |D_x|$. Write $\theta > 0$ for the relaxation time called τ in the new paper, avoiding confusion with the project's retarded time. Its equations (7.1)–(7.2) are

$$M_0 = -\frac{2k^2}{\omega}\mathcal{H}\partial_X |A|^2, \qquad \theta M_T + M = M_0. \tag{3.1}$$

For real forcing and real initial M_{in},

$$M(T) = e^{-T/\theta} M_{\mathrm{in}} + \theta^{-1} \int_0^T e^{-(T-s)/\theta} M_0(s)\,\mathrm{d}s \tag{3.2}$$

is real. An initial-value simulation must specify M_{in}; a steady harmonic transfer function omits this transient.

For time dependence $e^{-i\Omega T}$,

$$G(\Omega) = \frac{1}{1 - i\Omega\theta}, \qquad G(-\Omega) = \overline{G(\Omega)}.$$

Its low-frequency approximation is $1 + i\Omega\theta$. In physical time this corresponds to

$$M = M_0 - \theta\,\partial_T M_0 + O(\theta^2),$$

under appropriate temporal regularity and with the initial layer accounted for. It remains real. Replacing the entire real signal by $(1 + i\beta)M_0$, for a constant positive β, does not preserve the required conjugate symmetry. A selected complex Fourier amplitude can have such a phase lag; that does not establish equality of the full nonlinear models.

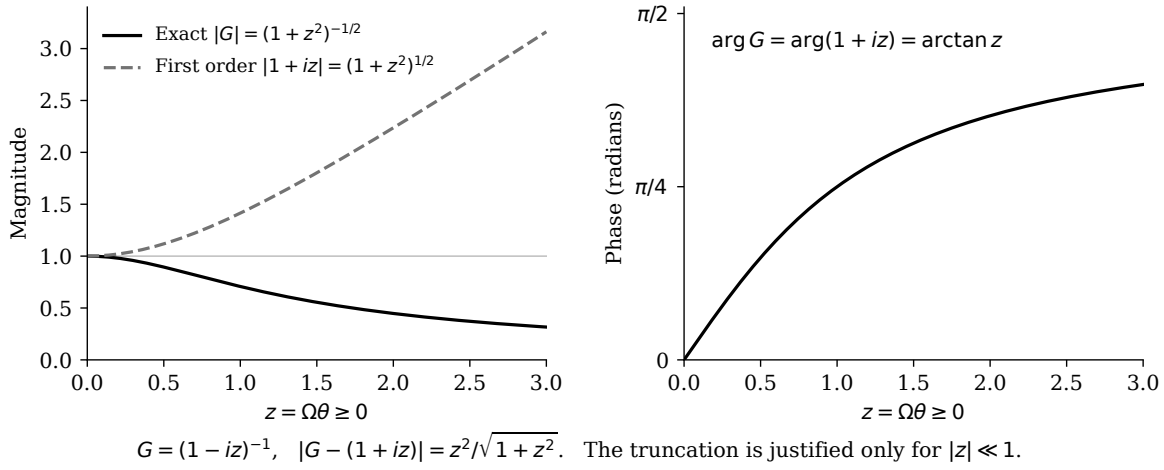


Figure 1: Analytic relaxation response for $z = \Omega\theta$. The exact and first-order transfer functions have the same phase, but different magnitudes. These are analytic curves, not simulated wave trajectories.

Indeed $G = (1 + iz)/(1 + z^2)$, so

$$|G - (1 + iz)| = \frac{z^2}{\sqrt{1 + z^2}}.$$

The assumption $\Omega\theta = O(\epsilon)$ places the correction one order beyond Dysthe; it is an imposed asymptotic ordering, not a measured adjustment time.

Proposition 3.1 (Mass balance for the complex-coefficient model)**.** *Let u be a sufficiently smooth L-periodic solution of* (2.1)*, with real ϵ, β. Set $\rho = |u|^2$, $Q(t) = \int_0^L \rho\,\mathrm{d}x$. Then*

$$Q'(t) = -4\epsilon\beta \int_0^L \rho\,|D_x|\rho\,\mathrm{d}x = -4\epsilon\beta \||D_x|^{1/2}\rho\|_{L^2}^2. \tag{3.3}$$

Proof. Differentiate Q to obtain $2\,\mathrm{Re}\int \bar{u}u_t$. The dispersive terms and the purely imaginary multiplication terms have zero real integral. The derivative nonlinearity gives a multiple of $\int \rho\rho_x = 0$. The remaining contribution to u_t is $-2\epsilon\beta u|D_x|\rho$. Parseval's identity gives the last equality because the multiplier $|k|$ is nonnegative. □

By contrast, a real M multiplying u with a real coefficient in the iu_t equation contributes zero to Q'. Replacing the real mean term of the $\beta = 0$ equation by such a relaxing M therefore preserves envelope mass. Additional viscous terms can change this conclusion; the comparison isolates the

mean-flow mechanism. Delay may affect phase and other energy exchanges, but it does not produce (3.3) through this substitution.

Even (3.3) does not prove global smoothness. It bounds mass and a time-integrated half derivative of *intensity*, not all derivatives of the envelope. For example, $|ae^{inx}|^2$ is constant although its envelope derivatives grow with n. This is a mathematical demonstration of missing control; arbitrarily large n violates narrowband physics.

4 Freeze the equation and the meaning of smoothness

The approved periodic spatial initial-value problem is

$$\begin{aligned} u_\xi = F(u) &= -iu_{\tau\tau} - i|u|^2u \\ &\quad - \epsilon\left(8|u|^2u_\tau + 2u^2\overline{u_\tau} + 2iu|D_\tau|(|u|^2)\right), \qquad (4.1) \\ u(0,\tau) &= u_0(\tau), \qquad \tau \in \mathbb{T}_L = \mathbb{R}/(L\mathbb{Z}). \qquad (4.2) \end{aligned}$$

Here ξ is propagation distance, τ retarded time, and u the scaled free-surface envelope. In the contract's carrier units, $\tau = \epsilon(2x - t)$, $\xi = \epsilon^2 x$, and $B = \epsilon u$. For $k_m = 2\pi m/L$, $|D_\tau|$ has multiplier $|k_m|$ and zero mean mode. The coefficient 2 of $u^2\overline{u_\tau}$ must not be replaced by the potential-envelope value 0. This is the local contract's Fedele–Dutykh spatial convention [3].

For $u = \sum_m a_m e^{ik_m\tau}$, set

$$\|u\|^2_{H^s(\mathbb{T}_L)} = L\sum_m (1 + k_m^2)^s |a_m|^2.$$

The intended finite-interval claim must specify a class, such as $C([0,\Xi]; H^s(\mathbb{T}_L))$. Well-posedness additionally requires uniqueness and continuous dependence in the stated spaces. The diagnostic choice $s = 3$ below is not claimed to be a known optimal existence threshold.

5 A numerical study of continued smoothness

The first computational question should be precise: for specified smooth initial packets, how far can solutions of (4.1) be followed with resolved derivatives and reproducible error estimates? A successful finite interval supplies evidence of continued smooth behavior on that interval. Extending the interval does not by itself establish global existence.

5.1 What can already be reused

The local `dysthe-core` reference evolves the approved spatial equation on a periodic profile interval of length L, with Fourier differentiation, retained modes $|m| < N/4$, and integrating-factor fourth-order Runge–Kutta steps in ξ. Its independent controls include the linear symbol, a nonzero wavenumber plane wave, the exact NLS traveling profile, and a direct Fourier convolution of the nonlinear terms including mean flow. The local `dysthe-experiments` code already supplies matched controls, phase/translation alignment, full periodic residual checks, and separate grid, step, and domain comparisons.

Stored results for $\epsilon = .025$, $L = 160$, $N = 4096$, $h = \Delta\xi = .00125$, and $0 \le \xi \le 4$ give the following maximum raw complex relative L^2 discrepancies. Each input was compared separately with $2N$, with $h/2$, and with $2L, 2N$ at fixed profile spacing.

Initial field	Grid	Step	Domain
Solitary profile	4.17×10^{-15}	3.07×10^{-9}	1.54×10^{-6}
Localized 2% perturbation	3.09×10^{-15}	4.08×10^{-9}	2.26×10^{-6}
Complex Gaussian	9.28×10^{-16}	4.05×10^{-11}	2.18×10^{-5}

These are existing two-level agreement checks, not rigorous error bounds or observed convergence orders. The stored qualification files were inspected for this review; these trajectories were not rerun. Fresh unit tests passed 21 of 21 core checks and 43 of 47 experiment checks; four HTTP tests could not open a socket in the execution sandbox. Passing tests neither qualifies a new packet nor proves a PDE theorem. The reference additions are local uncommitted work; a reproducible campaign must record their source hashes.

5.2 A bounded campaign with explicit stopping rules

Use $\epsilon \in \{0, .025, .05\}$, treating zero as the mathematical NLS control. Start with the existing smooth Gaussian and qualified traveling profiles, then add signed amplitude and phase perturbations outside the profile-construction symmetry. Keep all parameters, initial Fourier coefficients, and initial-condition identifiers fixed within a refinement comparison. The first task is a few documented cases, not a large training dataset.

Start at $(N, L, h) = (4096, 160, .00125)$. Compare $N, 2N, 4N$ at fixed L with a separately qualified step; compare $h, h/2, h/4$ at fixed fine grid; enlarge to $2L$ and, when necessary, $4L$ at fixed profile spacing. Reconstruct localized initial data on each domain rather than repeating its periodic copies. Recomputed solitary profiles must retain and report their initial-domain discrepancy. The nonlocal mean flow can change with L even when edge amplitudes are tiny.

Use successive horizons

$$\Xi \in \{1, 2, 4, 8, 16, 32\},$$

advancing only after the preceding horizon passes its checks. Compare all saved stations and inspect denser output near rapid growth. Retain raw complex errors as well as errors after fitting only translation and constant phase; never fit away amplitude changes. Perturbations should also be compared with their matched propagated control. A failed gate calls for refinement or a shorter qualified interval; it is not evidence of blow-up or nonexistence.

For $k_m = 2\pi m/L$ and Fourier coefficients normalized by $1/L$, record

$$\|u\|^2_{H^s(\mathbb{T}_L)} = L \sum_m (1 + k_m^2)^s |\widehat{u}_m|^2, \qquad s = 1, 2, 3. \tag{5.1}$$

Also record $\|u\|_\infty$, $\|u_\tau\|_\infty$, and spectral slopes above the resolved signal floor. Monitor the fraction of each H^s sum in the upper fifth of the retained band. A small unweighted spectral tail can conceal a large derivative error. Check boundary and exterior contributions in derivative norms, not action alone. Apparent norm growth is interpretable only while these weighted tails, slopes, and refinement comparisons remain resolved.

Provisional gates are maximum raw relative L^2 discrepancy 10^{-4}, H^1 discrepancy 10^{-3}, and H^2, H^3 discrepancies 10^{-2}, with upper-band weighted fractions below 10^{-6}. Use the finer field for normalization and absolute scales when the reference norm is near zero. For a weak perturbation, additionally require the signed response discrepancy below 2% of the measured response above its numerical floor. These are proposed acceptance tolerances, not established error bounds; tighten them when the scientific effect is smaller. Step halving must show the expected order before roundoff, rather than just two close answers. The existing $\epsilon\|u\|_\infty$ and bandwidth screens remain separate physical-model checks; their thresholds are not universal validity theorems.

Evaluate the unprojected RHS on an enlarged Fourier grid, retaining discarded nonlinear modes. For a continuous reconstruction v of a numerical trajectory, the useful defect is $r = v_\xi - F(v)$, including error from the reconstruction in ξ. The existing snapshot RHS diagnostic alone does not provide that time-dependent defect or its integral. Mass drift is an additional diagnostic; it cannot replace derivative resolution or residual control.

6 A Fourier-in-profile physics-informed network

6.1 Represent the periodic field and retain the full residual

The recommended first PINN is a neural approximation of the Fourier coefficients as functions of propagation distance:

$$v_\vartheta(\xi,\tau) = \sum_{m=-K}^{K} a_m(\xi;\vartheta)e^{ik_m\tau}, \qquad a_m \in \mathbb{C}. \tag{6.1}$$

A smooth network takes ξ and returns the real and imaginary parts of all coefficients. Periodicity is exact. Because the envelope is complex, *do not impose* $a_{-m} = \overline{a_m}$ or conjugate-reflection symmetry. Choose K from convergence of the reference fields in H^3, and repeat with a larger K; action capture alone is insufficient.

Automatic differentiation supplies a_m', while the profile derivatives use ik_m and $-k_m^2$. At each sampled ξ, form $\rho = |v_\vartheta|^2$, compute $|D_\tau|\rho$ with multiplier $|k_m|$ and zero mean mode, and evaluate every term of (4.1). Define

$$r_\vartheta = \partial_\xi v_\vartheta - F(v_\vartheta). \tag{6.2}$$

The cubic terms, including $v_\vartheta|D_\tau|(|v_\vartheta|^2)$, can contain modes through $3K$. The residual must retain these modes even though the trial field stops at K. A uniform quadrature grid with $Q > 6K$ points resolves this full residual and its squared L^2 norm for the trigonometric trial field. Penalizing only $|m| \le K$ could conceal a large discarded RHS and merely solve a truncated coefficient system. Independently validate with larger Q, larger K, and new propagation stations.

6.2 Hard initial values and short propagation windows

On $[\xi_j, \xi_{j+1}]$, write

$$a_m(\xi;\vartheta) = a_m^{(j)} + (\xi - \xi_j)b_m(\xi;\vartheta).$$

This enforces the supplied initial coefficients exactly. On the first window, report the projection error of the prescribed initial field onto $|m| \le K$; a hard condition does not remove this error. On later windows, set $a_m^{(j)}$ to the previous learned terminal value. Errors therefore accumulate honestly. Replacing every window's initial value with a reference snapshot defines a different, data-assisted experiment and must be reported separately.

Begin with short windows inside $\Xi = 1$, double precision, a smooth modest network, and at least three recorded initialization seeds. A practical training sequence is Adam followed by a bounded L-BFGS refinement. Declare the architecture, coefficient scaling, optimization budget, window length, and stopping rules before comparison. A basic loss is

$$\mathcal{L}_{\mathrm{PDE}} = \frac{1}{(\xi_{j+1} - \xi_j)L} \int_{\xi_j}^{\xi_{j+1}} \int_{\mathbb{T}_L} |r_\vartheta|^2 \, \mathrm{d}\tau \, \mathrm{d}\xi.$$

An explicitly stated Fourier-weighted residual may improve derivative control, but changes the training objective and must be tested. Conservation is a diagnostic rather than a substitute for the

PDE loss. Do not supply interior reference snapshots to a purported physics-only run; if such data are used, report that variant alongside it.

6.3 What would count as a meaningful result

First require the NLS and plane-wave controls, then the nonconstant mean-flow case, then the packet campaign. At held-out stations compare the network with the refined reference in raw complex L^2, H^1, H^2, H^3, peak and profile slope, phase where amplitude is meaningful, and full residual. Compare matched perturbation responses before interpreting their growth. Training failure, a small collocation loss, or disagreement between two networks cannot decide existence. A residual becomes an error certificate only when combined with justified stability estimates or validated inverse bounds of the kind discussed elsewhere in this report.

The active learning repository has no implemented water-wave PINN. This review adds no new trajectory or trained network. Its proposed first deliverable is one reproducible short-window comparison, followed by horizon extension only after qualification. Record total training and evaluation time, peak memory, all restarts and seeds, and any reference-generation cost; compare with the numerical solver at matched accuracy. A PINN has no presumed speed advantage for this one-coordinate problem. Its potential value must come from a demonstrated use—for example, a differentiable family of accurate approximations—rather than the name of the method.

7 What would establish continued smooth existence?

The question is whether prescribed smooth initial data generate a solution of (4.1) that continues to exist with controlled derivatives as the propagation distance increases. Three claims must be separated: existence on a specified finite interval; continuation for every finite distance; and smoothness of all orders. A computation over a finite interval can support the first claim empirically. Neither a bounded maximum amplitude nor a bounded L^2 norm controls all derivatives.

7.1 The available theory must match the equation

Mosincat, Pilod and Saut prove local well-posedness in $H^s(\mathbb{R}^2)$, $s > 0$, and global well-posedness and scattering for sufficiently small $L^2(\mathbb{R}^2)$ data for their *temporal*, nonlocal Dysthe equation; see their equation (1.12) and Theorems 1.1–1.2 [4]. Their one-dimensional whole-line discussion concerns temporal Dysthe with third-order dispersion and $H^s(\mathbb{R})$, $s \geq 1/4$. These statements do not directly cover our second-order spatial equation on a periodic profile domain. Nonzero constant backgrounds are also outside the finite-$L^2(\mathbb{R}^2)$ hypothesis.

For one-dimensional periodic temporal Dysthe, Heller and Shao obtain estimates and low-regularity ill-posedness results, but explicitly do not prove local well-posedness of the full equation [5, §1.5]. Belloni and Giuliani's recent global result for a local periodic NLS–Dysthe family likewise does not settle our problem: their discussion explicitly excludes the water-wave mean-flow term $u|D|(|u|^2)$ [6, pp. 5–6]. Thus we have not identified a theorem whose stated hypotheses match (4.1). This is a limitation of the present review, not a claim that all Dysthe existence questions are open.

7.2 A rigorous finite-dimensional statement, and its limit

Let $\mathbb{T}_L$ be the periodic profile circle of length $L > 0$, and let P_K denote the orthogonal Fourier projection onto modes $|n| \leq K$. Write $F(u)$ for the full right-hand side of (4.1). The following

statement concerns exact Galerkin evolution, with exact nonlinear products and the stated Fourier mean-flow operator.

Proposition 7.1 (Global existence at each fixed Fourier cutoff)**.** *Let K be a nonnegative integer, let $\epsilon \in \mathbb{R}$, and let $u_K(0)$ belong to the range of P_K. The finite system*

$$\partial_\xi u_K = P_K F(u_K)$$

has a unique solution for every $\xi \in \mathbb{R}$, and $\|u_K(\xi)\|_{L^2(\mathbb{T}_L)} = \|u_K(0)\|_{L^2(\mathbb{T}_L)}$.

Proof. Its Fourier coefficients solve a polynomial ordinary differential equation in finitely many real variables, hence have a unique local solution. Put $\rho = |u_K|^2$. Self-adjointness of P_K and $P_K u_K = u_K$ allow its removal from the mass derivative. The linear dispersive term vanishes after integration by parts. The local cubic term and the mean-flow term have zero real contribution because ρ and $|D_\tau|\rho$ are real. The remaining two derivative terms give

$$\frac{\mathrm{d}}{\mathrm{d}\xi}\|u_K\|_{L^2}^2 = 2\,\mathrm{Re}\int_{\mathbb{T}_L} \overline{u_K} F(u_K)\ \mathrm{d}\tau = -10\epsilon \int_{\mathbb{T}_L} \rho\,\partial_\tau \rho\ \mathrm{d}\tau = 0.$$

Parseval's identity therefore bounds the coefficient vector throughout its maximal interval. In finite dimension it remains in a compact ball, where the polynomial vector field and its derivative are bounded. The ordinary differential equation continuation theorem excludes a finite endpoint in either direction. □

This proposition does not prove existence for the partial differential equation, or stability of a chosen time-stepping method. For $s > 0$ it gives only

$$\|u_K(\xi)\|_{H^s(\mathbb{T}_L)} \le \left(1 + \left(\frac{2\pi K}{L}\right)^2\right)^{s/2} \|u_K(0)\|_{L^2(\mathbb{T}_L)},$$

whose constant grows with K. Global finite systems can therefore fail to provide the uniform regularity needed in the limit $K \to \infty$. An aliased implementation need not equal this Galerkin system, and an RK4 update does not inherit its exact mass conservation or global bound.

7.3 The analytic continuation route

A PDE proof must first establish local existence, uniqueness and continuous dependence in a specified function space for the exact equation and boundary conditions. Directly applying a finite-dimensional ODE argument is insufficient: the nonlinear terms contain derivatives, including the first-order nonlocal operator $|D_\tau|$, so a naive same-H^s nonlinear estimate loses derivatives. Suitable cancellations, modified energy estimates, or dispersive estimates must recover this loss.

If a local theory supplies a lifespan bounded below in terms of the H^s norm, its restart argument gives a continuation criterion: a finite maximal distance can occur only if that controlling norm becomes unbounded. Establishing the needed norm bound uniformly on every finite interval would then give global existence in that regularity class. Passing from Galerkin approximations also requires sufficient compactness, convergence of the derivative nonlinearities, and uniqueness of the limit. For initially smooth data, a claim of smoothness of all orders additionally needs persistence of higher regularity on the same interval; one fixed H^s estimate alone is not that claim.

Residual-based PINN error estimates also require PDE stability [7, 8]. For an exact solution u and a candidate v, put $E(\xi) = \|u(\xi) - v(\xi)\|_{H^s}$ and $r = \partial_\xi v - F(v)$. If a justified stability argument

gives $E' \le CE + \|r\|_{\mathcal{R}}$ almost everywhere, with $C \ge 0$ and an appropriate residual norm $\mathcal{R}$, then Gronwall's inequality gives

$$E(\xi) \le e^{C\xi}\left(E(0) + \int_0^{\xi} \|r(\eta)\|_{\mathcal{R}}\ \mathrm{d}\eta\right).$$

For this equation, derivative loss must actually be controlled and C must be bounded on the relevant solution set, independently of a numerical cutoff. Without that work, the displayed bound is conditional; an error estimate presupposing an exact solution is not an existence proof.

7.4 A rigorous route from a computed candidate

A numerical solution or PINN can instead serve as the center of an *a posteriori* existence proof, following the general validated-numerics approach [9]. On a fixed interval $[0, \Xi]$, choose Banach spaces X, Y and an initial-value defect map, for example

$$\mathcal{G}(v) = (\partial_\xi v - F(v),\ v(0) - u_0).$$

The spaces must make this map well-defined and must encode periodicity and the desired regularity. Complex-valued functions are treated as real Banach spaces because the equation contains complex conjugation.

Theorem 7.2 (Conditional contraction certificate)**.** *Let $U \subseteq X$ be open, let $\mathcal{G} : U \to Y$ be continuously Fréchet differentiable, and let $A : Y \to X$ be bounded, real-linear and injective. Suppose $\overline{B}_X(\widetilde{u}, r) \subseteq U$, $r > 0$, and verified bounds satisfy*

$$\|A\mathcal{G}(\widetilde{u})\|_X \le Y_0, \qquad \sup_{v \in \overline{B}_X(\widetilde{u}, r)} \|I_X - AD\mathcal{G}(v)\|_{X \to X} \le Z(r).$$

If $Z(r) < 1$ and $Y_0 + Z(r)r \le r$, then $\mathcal{G}$ has exactly one zero in this closed ball.

Proof. Define $T(v) = v - A\mathcal{G}(v)$. The derivative bound and the mean-value inequality show that T is Lipschitz on the ball with constant $Z(r)$. For v in the ball, $\|T(v) - \widetilde{u}\|_X \le Z(r)r + Y_0 \le r$. The contraction theorem gives a unique fixed point. Injectivity of A makes a fixed point equivalent to a zero of $\mathcal{G}$, proving both existence and uniqueness within the ball. □

This is a sufficient criterion, not a certificate already obtained for our equation. Constructing X, Y, A and proving the displayed bounds is the main mathematical work. The validation must cover all unresolved Fourier modes, the nonlocal operator, roundoff with outward bounds, and the initial condition including any initial tail. For successive slabs, it must carry the certified endpoint uncertainty forward. A residual sampled on a grid, or at PINN training points, supplies none of those uniform guarantees by itself. Such a certificate establishes the specified finite interval; global continuation still requires control over arbitrarily long intervals.

8 Recommended sequence and limits of the present review

First freeze the equation and domain and map candidate existence theorems term by term. Next extend the existing Gaussian benchmark with derivative histories, three-level checks, and explicit stopping reasons. Then build one short-window spectral PINN, test multiple windows and seeds, and measure full residuals and total cost. Finally attempt one finite-interval certificate, identifying precisely which estimate remains missing if certification fails.

A later damping study would require separately versioned temporal constant-β and real-relaxation models, with specified initial mean flow. Test their different mass laws before interpreting either as a smoothness mechanism. Treat small relaxation times implicitly or exponentially. Do not relabel either model as spatial-v1.

The defensible first research claim concerns resolved finite-distance smooth evolution and independently tested neural approximations for a specified water-wave model. A rigorous local certificate is a stronger second result. Global smoothness, physical wave breaking, and first-principles nonlinear damping require further arguments.

Verification status. The review used the supplied PDFs, visual checks of mathematical pages, current local scope contracts and code, stored qualification records, fresh unit tests, and primary literature checked as of 23 October 2026. Original PDFs and active repository files were not modified. Some inspected reference additions are uncommitted or untracked; local capability is not a claim about a published release. The learning registry has no implemented active water-wave PINN, and the paper repository's claims file still records no new evidence.

The portable source package includes editable LaTeX, style, the vector PDF figure, independent identity checks and results, build instructions, and verification metadata. Five low-mode test profiles verified the spatial mass balance to 3.1×10^{-16}, the complex-coefficient mass balance to 2.7×10^{-16}, and real-mean-flow mass preservation to 4.0×10^{-16} in absolute discrepancy. These floating-point checks support the explicit algebra; they are not interval-arithmetic certificates or new evolution results. The full fifth-order harmonic derivation was not reproduced. No new PINN, continuation dataset, existence certificate, or laboratory comparison was produced in this review.

References

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